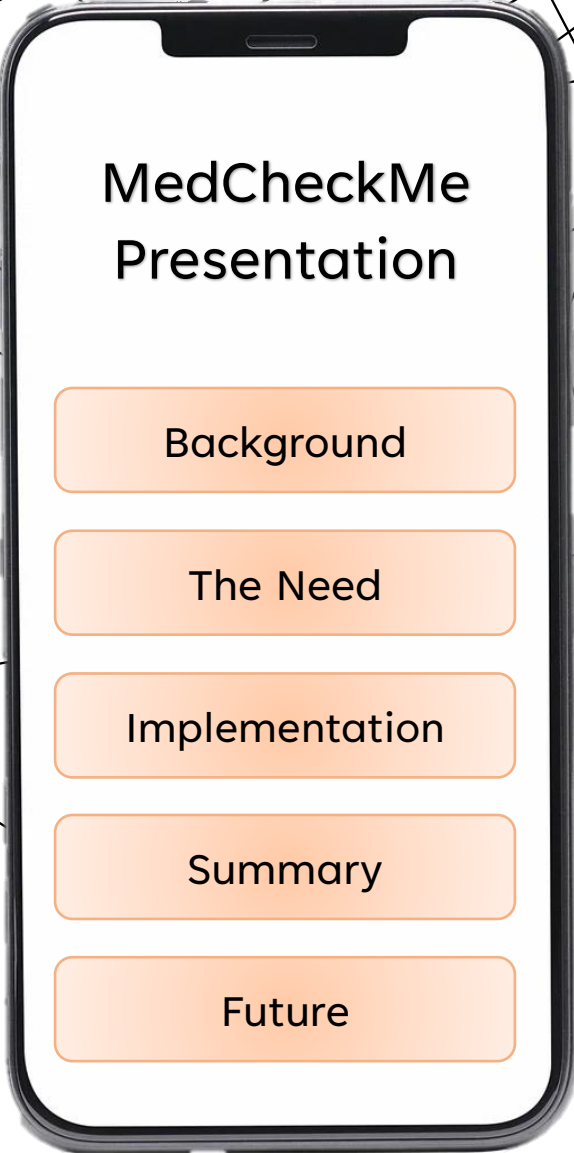




MedCheckMe Presentation

Background

The Need

Implementation

Summary

Future

MEDCHECKME APP

Sponsor: Marc Blumenthal
Social Ventures Foundation

Advisor: Miguel Mireles-Nunez

Team:

Justus Morales

Ryan Potechin

Sophie Chabot

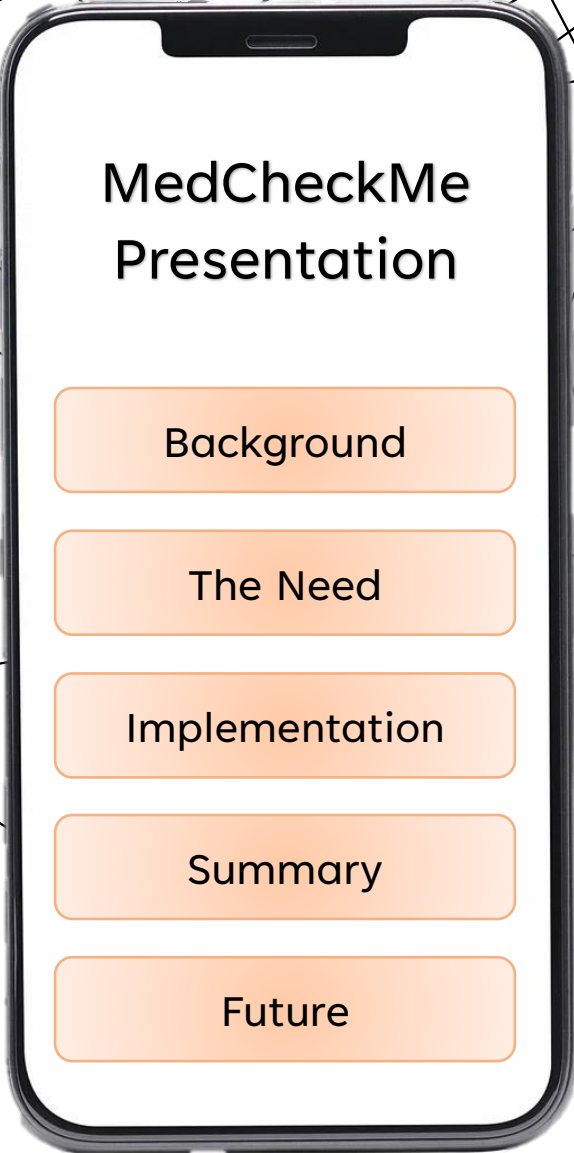
Samantha Lee

Thomas Young

Trevor Lau

18 October 2023

BIOE4792 Bioengineering Capstone 2



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BIOE4792 Bioengineering Capstone 2



DISEASE HAS A LARGE IMPACT ON LOW-INCOME COMMUNITIES



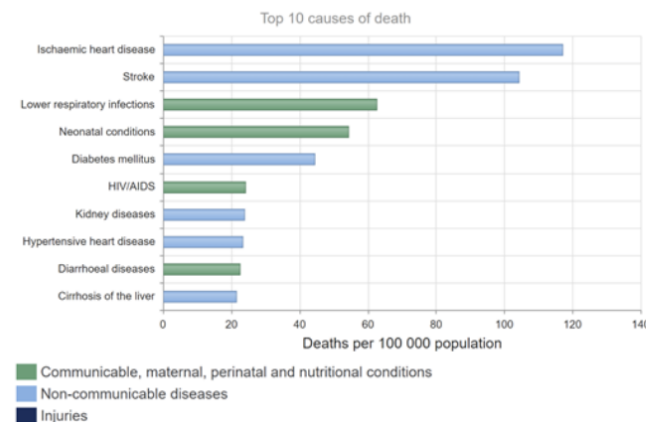
Lack of access to medical checkups in developing countries has a significant impact on global health.



Upwards of 3.5 billion people lack access to necessary healthcare services, aims to address this issue with a multi-purpose application for health monitoring.

Top 10 causes of death in Haiti for both sexes aged all ages (2019)

[Show filters](#) | [Top-10 deaths](#) | [Top-10 DALYs](#) | [Underlying data](#) | [Download with OData API](#)



Early detection can...

- prevent fatalities from non-communicable diseases by managing symptoms early
- prevent spread of communicable diseases by providing proper curative treatment early



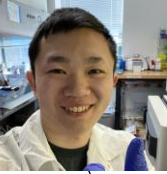
PROVIDING LOW COST HEALTHCARE TO LOW INCOME COMMUNITIES

Social Ventures Foundation & Micro franchising

- Non-profit organization supporting social entrepreneurs
- Inspires global movement for positive social change
- Small cost to start, operations can be replicated easily
- Provides economic development
- Creates jobs to help communities step out of poverty

MedCheckMe App

- An app to provide affordable healthcare and social impacts to lower income communities
- Provides preliminary tests for diseases affecting:
Eye, Skin, Hearing, Heart Rate, Blood Pressure, Respiration Rate
- Includes unique patient identifier

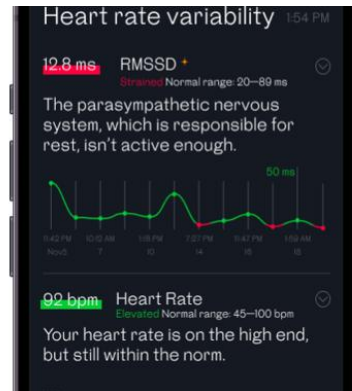


EXISTING SOLUTIONS TEND TO FALL SHORT.



MDLIVE

Contactless
appointments



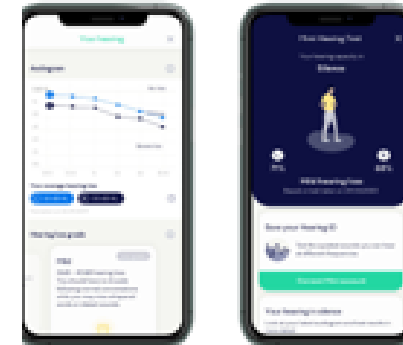
Welltory

Records PPG data



DermLite HUD

Dermatoscope attach
ment



Mimi Hearing App

At-home hearing test



*In 2023, poverty likely to increase
to 63% of Haitian population
below \$3.65/day
-The World Bank*

Diagnostic check-ups
shall be kept below \$1

*As of 2021, only 39% of the
Haitian population used the
internet
-The World Bank*

App shall not require
constant internet
connection

DESIGN REQUIREMENTS

App shall provide necessary diagnostic
tests conforming to regulatory standards



ISO 81060



ISO 8253



WE CONSIDERED VARIOUS DIFFERENT DIAGNOSTIC EXAMS

Skin

- Lesions
- Fungal Infections
- Rashes

Ears

- Poor Hearing
- Physical Obstruction

Cardiovascular

- High Blood Pressure
- Heart Rate Irregularities

Eyes

- Colorblindness
- Poor Vision
- Retinal Diseases

Respiratory

- Irregular Lung Sounds
- Respiratory Diseases

Urinalysis

- Various ailments with measurable affects found in urine



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Cardiovascular

- High Blood Pressure
- Heart Rate Irregularities

Eyes

- Colorblindness
- Poor Vision
- Retinal Diseases

Decision was made with decision matrix scoring the (1) relevance of data, (2) ease of measurement, (3) precedent for phone-based accuracy, and (4) feasibility within capstone scope.



WE CONSIDERED VARIOUS PHONES TO HOST THE PROTOTYPE

Criteria	Cost	Battery Life	Camera	Storage	Relevance	Score
Weight	35	15	25	10	10	
Samsung Galaxy S20	3.61	2.26	2	5	3.72	2.97
Samsung Galaxy S21 5G	2.63	2.53	4	5	3.72	3.17
Samsung Galaxy S22	2.49	2.51	4	5	3.72	3.12
Samsung Galaxy S23	0.85	4.40	5	5	3.72	3.08
ASUS ROG Phone 5S	2.82	3.94	3	5	3.72	3.20
ASUS ROG Phone 6	1.97	4.85	5	5	3.72	3.41
Samsung Galaxy a32	4.00	4.52	3	3	3.72	3.50
Samsung Galaxy A54	3.80	5.00	3	5	3.72	3.70
iPhone 8	4.85	1.90	1	3	0.62	2.60
iPhone X	4.79	1.85	2	3	0.62	2.81
iPhone 11	2.49	4.47	3	3	0.62	2.65
iPhone 12	1.84	3.97	4	3	0.62	2.60
iPhone 13	1.51	2.63	5	5	0.62	2.61
Pixel 7	2.16	3.26	5	5	3.72	3.37
Pixel 6	3.80	3.45	5	5	3.72	3.97
Pixel 5	4.92	3.79	3	5	3.72	3.91
Pixel 4	5.00	1.00	2	3	3.72	3.07



- After quantitatively ranking relevant aspects of various phone models and operating systems, we chose Android OS and the Google Pixel 5 as the pilot device.



HEARING LEVELS ARE DETERMINED USING TWO PRIMARY FORMS OF PURE-TONE TESTING

Pure-tone testing is used to test the functioning of the outer and middle ear.¹

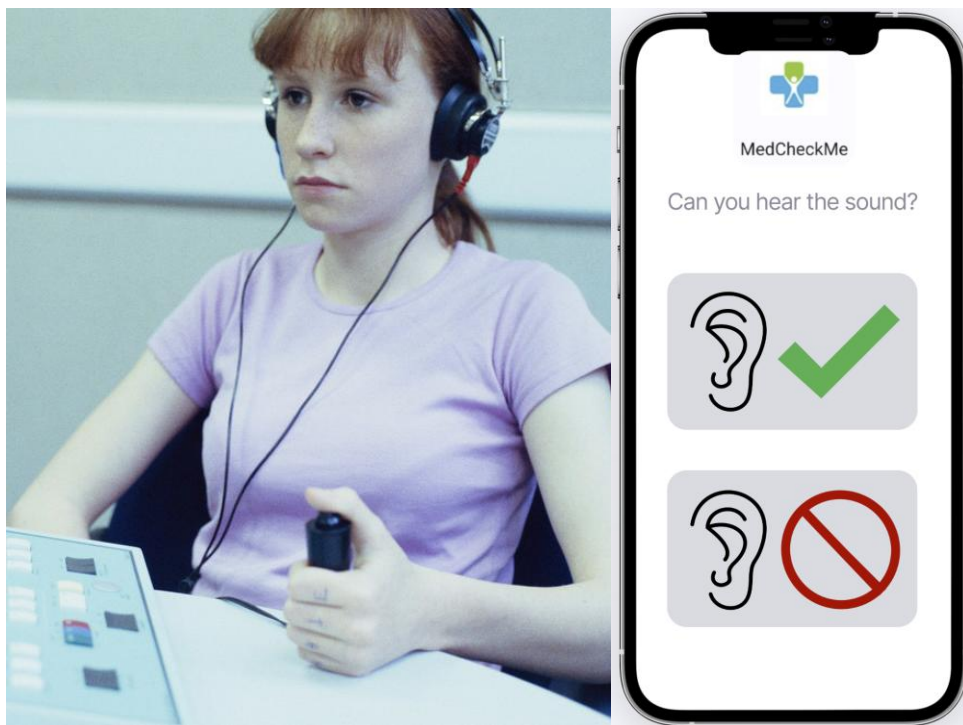


Photo credit: Science Photo Library

[1] <https://www.asha.org/public/hearing/pure-tone-testing/>

• Pure-Tone (PT) Testing Method

- Playback of tone in various decibels (0-120 dB)
- Record lowest decibel of that frequency audible to patient
- Repeat for each frequency from 250-8000 Hz
- Display audiogram graph for analysis

• Masking Method for Pure-Tone Testing

- Repeat PT Testing Methods for a single ear
- Record lowest audible decibel per frequency for each isolated ear
- Display data for left and right ear on audiogram



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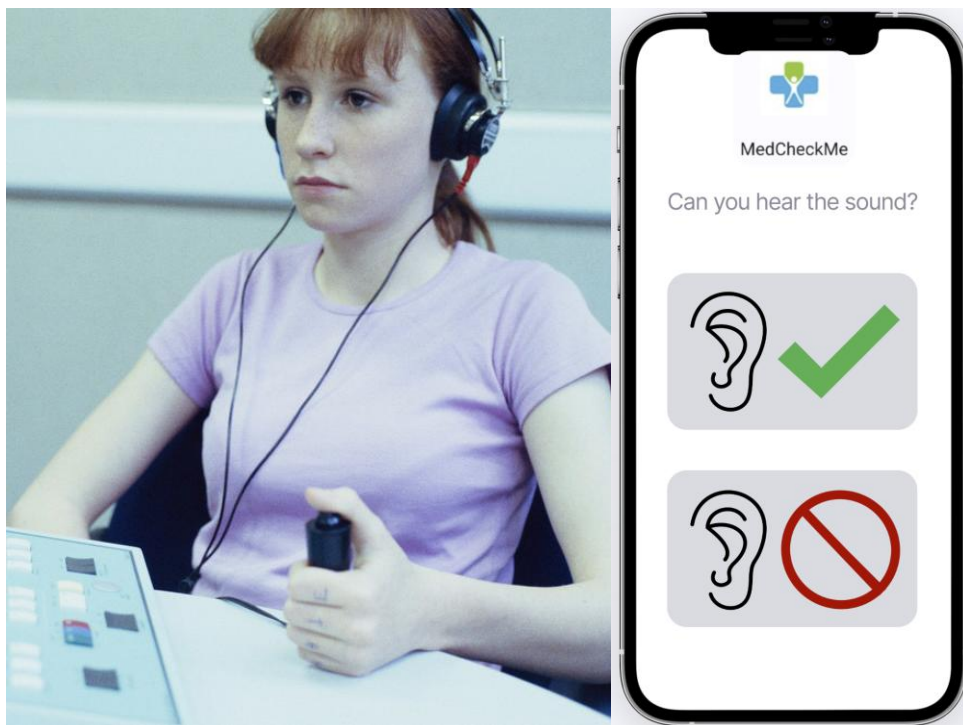


Photo credit: Science Photo Library

[1] <https://www.asha.org/public/hearing/pure-tone-testing/>

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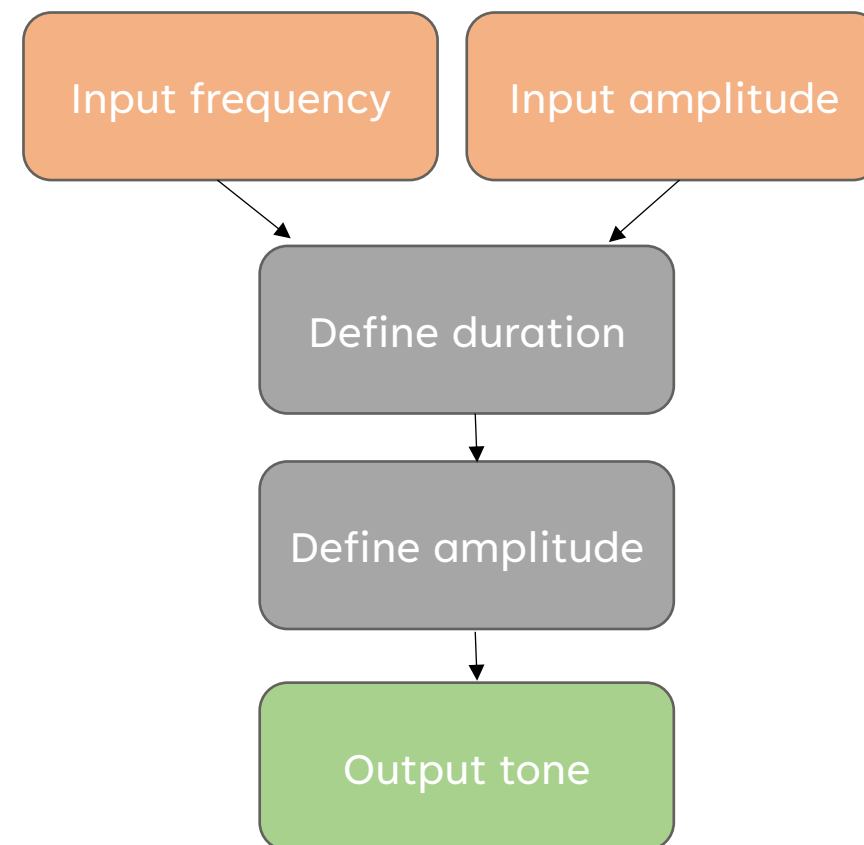
- Repeat PT Testing Methods for a single ear
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- Display data for left and right ear on audiogram



MATLAB CODE AND RESULTS FOR PT TESTING

GenSignal Function: to generate a signal at a specified frequency and decibel for playback

```
GenSignal.m  x  +
1  function [signal] = GenSignal(frequency,amplitude_dB)
2
3  % Define the parameters
4  duration = 1;    % Duration in seconds
5
6  % Calculate the amplitude in linear scale from dB
7  amplitude = 10^(amplitude_dB/20);
8
9  % Create a time vector
10 t = linspace(0, duration, duration * 44100); % Assuming a 44.1 kHz sample rate
11
12 % Generate the tone
13 signal = amplitude * sin(2 * pi * frequency * t);
14
15 end
```





MATLAB CODE AND RESULTS FOR PT TESTING

PT Test: calls GenSignal function to create sound at specified frequency and decibel then prompts user feedback and stores lowest recorded decibel.

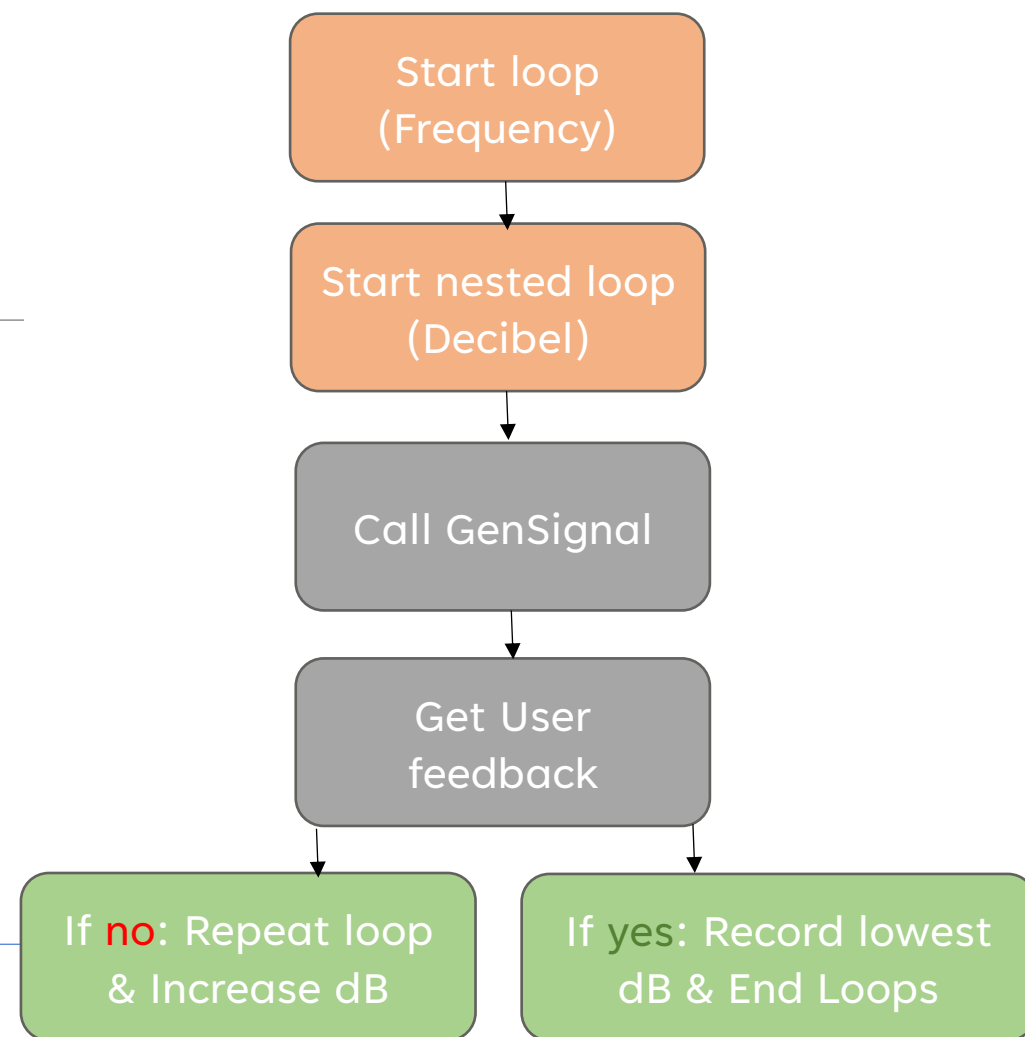
```
% Test for both ears
for frequency = [250,500,1000,2000,4000,8000]
    for amplitude_dB = 3:5:120
        signal = GenSignal(frequency,amplitude_dB);

        sound(signal, 44100) % Adjust the sample rate as needed

        % Ask for user input if sound playback was audible
        userInput = input('Can you hear the sound? Enter yes or no: ', 's');

        % Display the user's input
        fprintf('You entered: %s\n', userInput);

        % End test once user can hear the decibel level and record data
        % into array
        if userInput == "yes"
            freq(end+1) = frequency;
            dB(end+1) = amplitude_dB - 5; % integer to subtract = integer used to increment
            break
        end
    end
end
```





MATLAB CODE AND RESULTS FOR PT TESTING

Masking Method for PT Test: Repeat same code for testing the left and right ear, except isolating one column in signal array for the corresponding side

Left ear

```
%% Test for left ear
for frequency = [250,500,1000,2000,4000,8000]
    for amplitude_dB = 3:5:120
        signal = GenSignal(frequency,amplitude_dB);

        % Play the sound
        sL = zeros(size(signal,2),2);
        sL(:,1) = signal.';

        sound(sL, 44100) % Adjust the sample rate as needed

        % Ask for user input if sound playback was audible
        userInput = input('Can you hear the sound? Enter yes or no: ', 's');

        % Display the user's input
        fprintf('You entered: %s\n', userInput);

        % End test once user can hear the decibel level and record data
        % into array
        if userInput == "yes"
            freq_L(end+1) = frequency;
            dB_L(end+1) = amplitude_dB - 5; % integer to subtract = integer used to increment
            break
        end
    end
end
```

Right ear

```
%% Test for right ear
for frequency = [250,500,1000,2000,4000,8000]
    for amplitude_dB = 3:5:120
        signal = GenSignal(frequency,amplitude_dB);

        % Play the sound
        sL = zeros(size(signal,2),2);
        sL(:,2) = signal.';

        sound(sL, 44100) % Adjust the sample rate as needed

        % Ask for user input if sound playback was audible
        userInput = input('Can you hear the sound? Enter yes or no: ', 's');

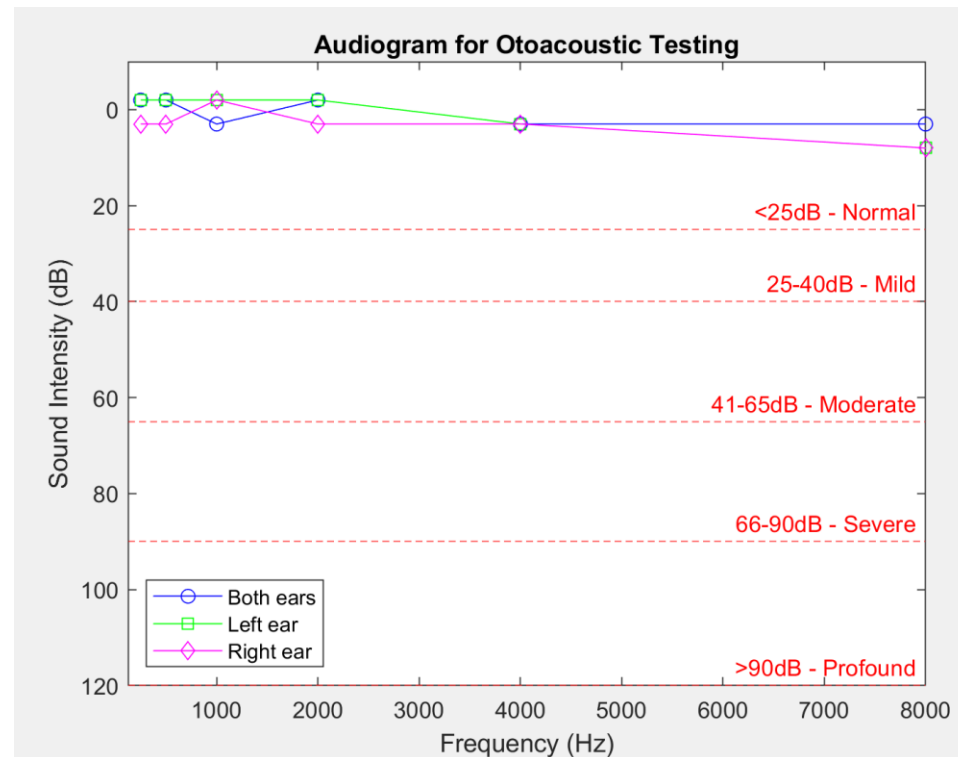
        % Display the user's input
        fprintf('You entered: %s\n', userInput);

        % End test once user can hear the decibel level and record data
        % into array
        if userInput == "yes"
            freq_R(end+1) = frequency;
            dB_R(end+1) = amplitude_dB - 5; % integer to subtract = integer used to increment
            break
        end
    end
end
```




MATLAB CODE AND RESULTS FOR PT TESTING

Generate audiogram: Shows audiogram graph with analytic guidelines for easy diagnosis



%% plot audiogram graph using data collected above

```
both = plot(freq,dB,'-ob');
hold on
left = plot(freq_L,dB_L,'-square');
right = plot(freq_R,dB_R,'-diamond');
xlim([125,8000])
ylim([-10,120])
xlabel('Frequency (Hz)')
ylabel('Sound Intensity (dB)')
title('Audiogram for Otoacoustic Testing')
set(gca,'YDir','reverse')

% Plot boundaries for analysis
yline([25,40,65,90,120],'--r',{'<25dB - Normal','25-40dB - Mild','41-65dB - Moderate','66-90dB - Severe','>90dB - Profound'})

% Create a legend
legend([both,left,right],'Both ears','Left ear','Right ear','','','','','Location','southwest')

return
```

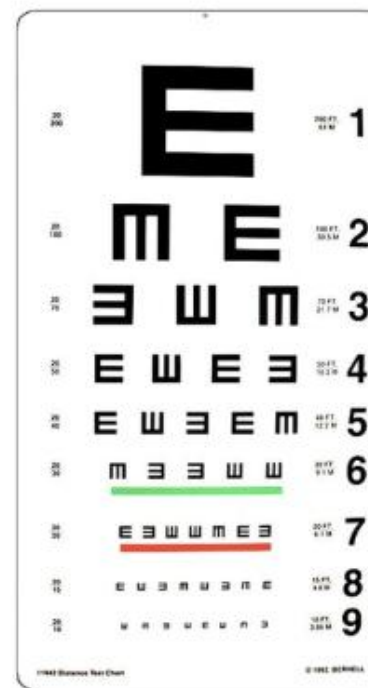



VISION ASSESSMENTS USING PHONE SCREEN

VISUAL ACUITY

Visual Acuity is a measurement of clarity and sharpness of one's vision and provides information for diagnosing and managing eye diseases.

- Tumbling E chart adapted to phone screen
- E pointing in different orientations (up, down, left, right)
- Covers one eye at a time, stands 2-3 meters away and points in direction of E.
- Adapted to phone screen (right)
- Test administrator swipes in the direction of E

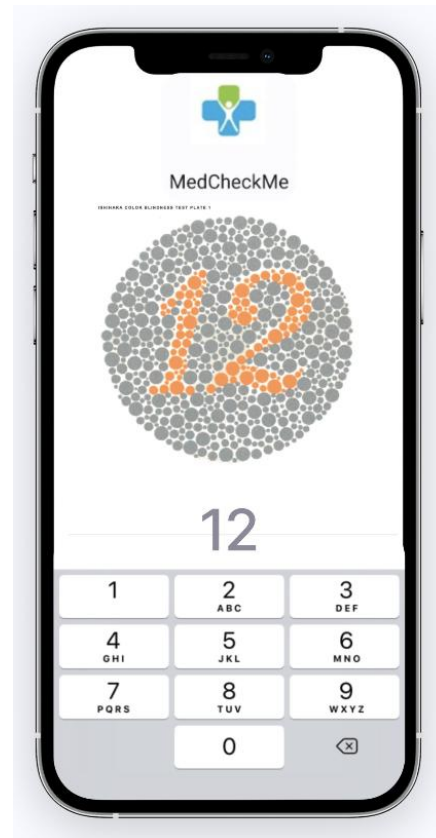


Tumbling E chart





VISION ASSESSMENTS USING PHONE SCREEN



ISHIHARA COLOR BLINDNESS PLATES

Ishihara color blind plates measure color deficiency of the congenital origin.

- Measuring Red-green deficiency, Protanopia or protanomaly, and Deuteranopia or deuteranomaly
- Phone held 75 cm from patient, shown one at a time, patient inputs number using keypad (right).
- 11 plates; 10 or more correct is normal vision, less than 7 is color deficiency



MATLAB SCRIPT FOR VISION TESTING

```
%Show plate 1
imshow("Plate1.png")

%Ask user for input
prompt="What number is shown?";

%Count
if input(prompt) == 12
    correct=correct+1;
else
    correct=0;
end

fprintf("Number correct " + correct + " out of 20")
fprintf("Number of Red Green Deficiencies: " + redgreen)
fprintf("Number of Deuteranopia Deficiencies: " + deut)
fprintf("Number of protanopia Deficiencies: " + pro)
```

Visual Acuity Test

- Varying E's shown one at a time
- Swipe up, down, left, or right



```
close all; clear vars; clc;

%Show E up
correct=0;
incorrect=0;

imshow("UpE.png")

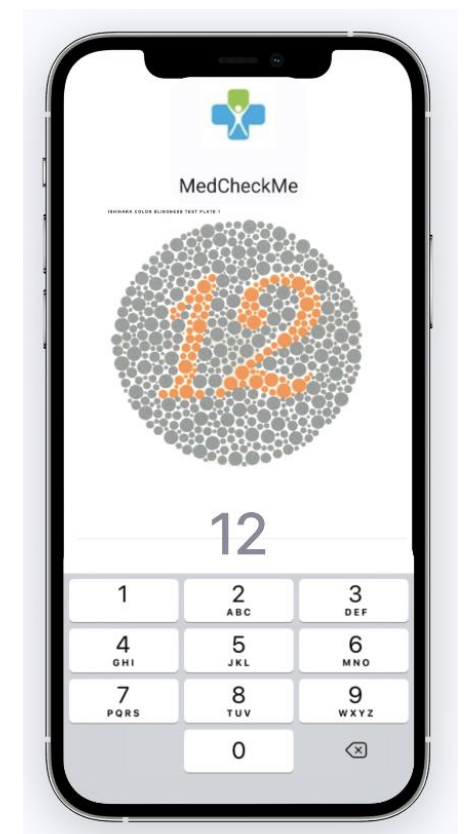
prompt="Which way is the E facing?"

k = waitforbuttonpress;
% 28 leftarrow
% 29 rightarrow
% 30 uparrow
% 31 downarrow
value = double(get(gcf,'CurrentCharacter'));

if value == 30
    correct=correct+1;
else
    incorrect=incorrect+1;
end
```

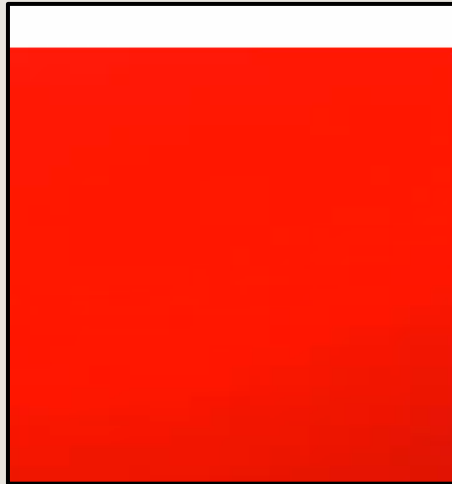
Ishihara Color Blind Test

- Plates shown one at a time
- Keypad used to type in answer



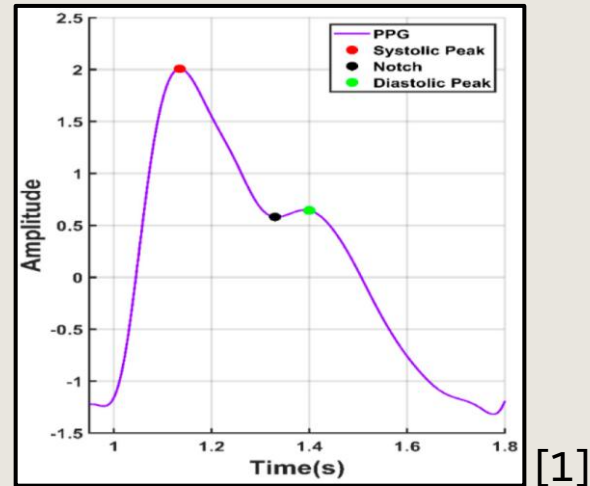


PPG DATA CAN BE COLLECTED FROM PHONE VIDEOS.



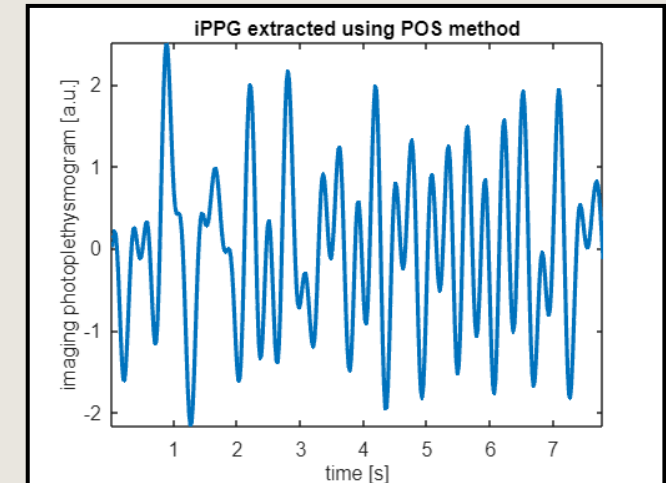
Analysis of Raw Video

- Videos are broken down frame-by-frame
- Light intensity data is recorded [2]



Filtering of Data

- Data is filtered using bandpass filters
- Removes wander and noise

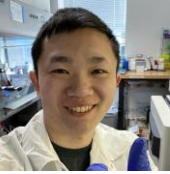


Production of Waveform

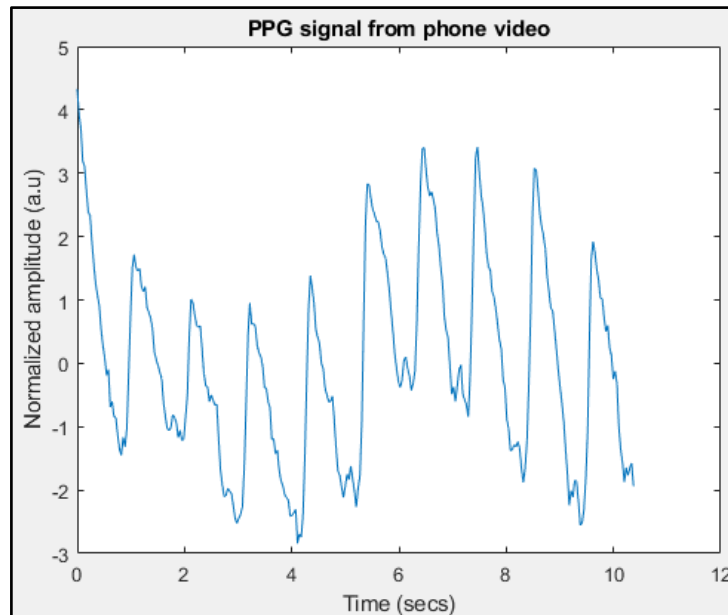
- Resulting filtered data is plotted against time

[1] M. H. Chowdhury *et al.*, *Sensors*, vol. 20, no. 11, p. 3127, 2020. doi:10.3390/s20113127

[2] A. M. Unakafov, "Pulse rate estimation using imaging photoplethysmography: Generic Framework and comparison of methods on a publicly available dataset," *Biomedical Physics & Engineering Express*, vol. 4, no. 4, p. 045001, 2018. doi:10.1088/2057-1976/aabd09



PULSE RATE CAN BE DETERMINED FROM PPG SIGNALS



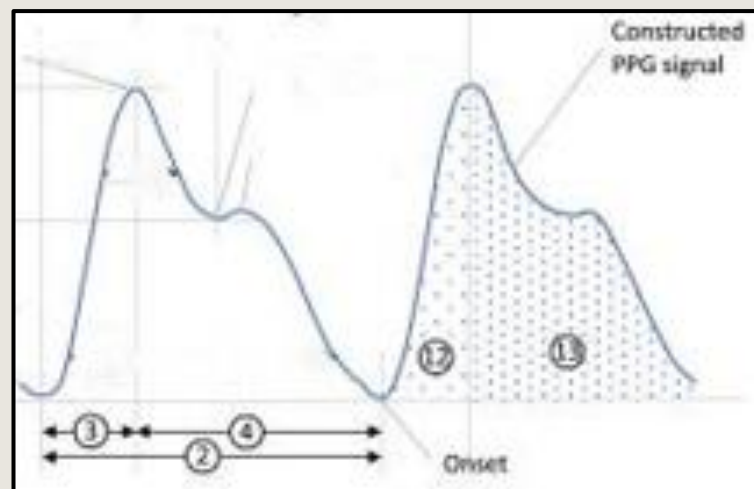
Example of PPG Signal
Extracted from Video

- Limits the extent of interfering factors during signal extraction
 - Spatial detrending
 - Bandpass filters
 - Outlier suppression
- Utilizes a combination of functions for calculation
 - Wavelet transforms
 - Discrete Fourier transforms



BLOOD PRESSURE CAN BE ESTIMATED FROM PPG DATA.

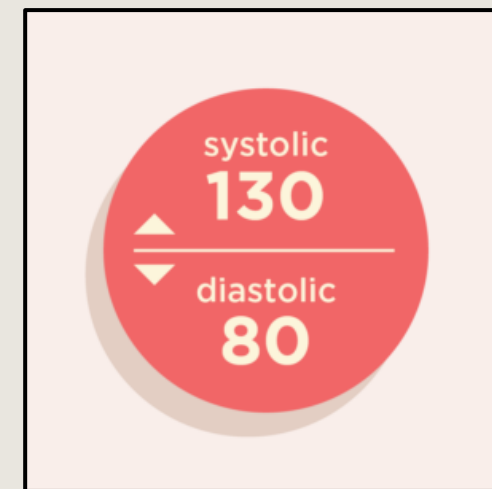
- **Pulse Wave Analysis Method**
 - Uses the PPG waveform to produce a BP estimate.
 - Based on waveform shape and features.
- **BP_annotate [1]**
 - Feature detection algorithm for arterial blood pressure
 - Takes in PPG waveform and returns the onset, systolic peak, dicrotic notch, and diastolic peak



[2]

Extraction of Important Features

- Important features are used in the estimation of BP
- Important features include Onset, Systolic Peak, Dicrotic Notch, and Diastolic Peak



[3]

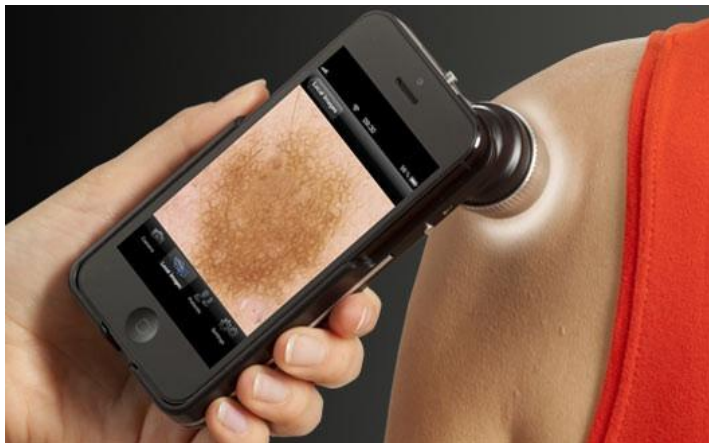
Estimation of Blood Pressure Values

- Feature set is fit to corresponding SBP and DBP value based on loaded dataset.
- Most studies using the PWA method have used the MIMIC II dataset.

[1] Alexandre Laurin (2023). BP_annotate (https://www.mathworks.com/matlabcentral/fileexchange/60172-bp_annotate), MATLAB Central File Exchange. Retrieved October 19, 2023.
 [2] L. Frey, C. Menon, and M. Elgendi, *npj Digital Medicine*, vol. 5, no. 1, 2022. doi:10.1038/s41746-022-00629-2
 [3] [1] V. Pelley, The Paper Gown, <https://www.zocdoc.com/blog/why-does-the-definition-of-normal-blood-pressure-keep-changing/> (accessed Aug. 16, 2023).

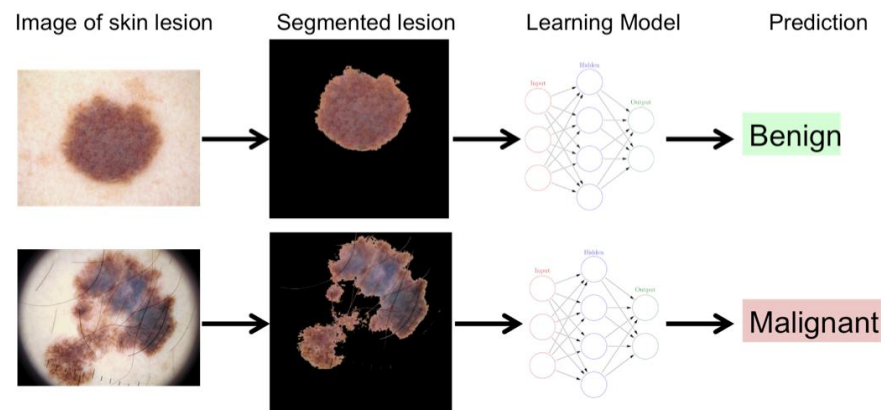


SKIN TEST CAN BE CONDUCTED USING A PHONE CAMERA



Phone Dermoscopy

- Dermatoscope attachment to phone camera can be used for imaging of skin lesions
- Research has proven phone dermoscopy to have similar fidelity to clinical grade dermascopes [1]



Machine/Deep Learning Algorithm Diagnosis

- A.I can be utilized to analyze features of an image of a skin lesion through a variety of different Learning Model approaches
- Previous use of Machine Learning algorithms for dermoscopic diagnosis have proven effective with high degrees of certainty [2]

Sources: [1] A. Yilmaz, G. Gencoglan, R. Varol, A. A. Demircali, M. Keshavarz, and H. Uvet, "MobileSkin: Classification of Skin Lesion Images Acquired Using Mobile Phone-Attached Hand-Held Dermoscopes," *Journal of Clinical Medicine*, vol. 11, no. 17, p. 5102, Jan. 2022, doi: <https://doi.org/10.3390/jcm11175102>.

[2]: M. Phillips et al., "Assessment of Accuracy of an Artificial Intelligence Algorithm to Detect Melanoma in Images of Skin Lesions," *JAMA Network Open*, vol. 2, no. 10, p. e1913436, Oct. 2019, doi: <https://doi.org/10.1001/jamanetworkopen.2019.13436>.



SKIN TESTING IMPLEMENTATION: DERMOSCOPE

Principle of Non-Polarized Dermoscopy

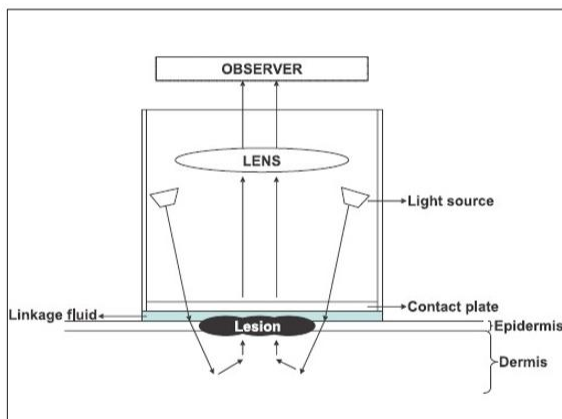


Figure 1: Optics of dermoscope. The refracted light transilluminates the lesion while passing through it and is perceived as a distinct pattern.

- A simple dermoscope can be made using a 10x magnification lens, a glass contact plate, and some type of light source.

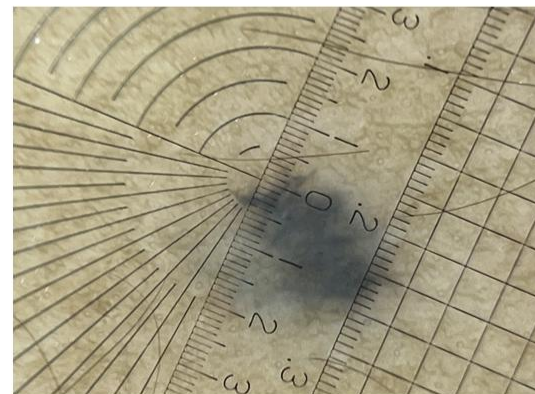
Imaging Solution



Illuminated Jeweler's Loupe (10x)

- Casing, faceplate, and lighting included
- Compact, easy to place over phone camera lens

Test Images Taken Using Device





SKIN TEST: IMAGE ANALYSIS

Image Processing Pipeline

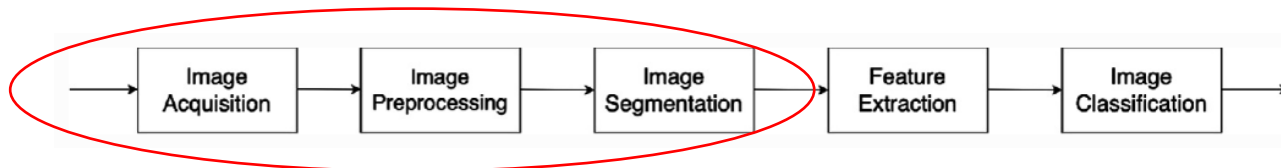
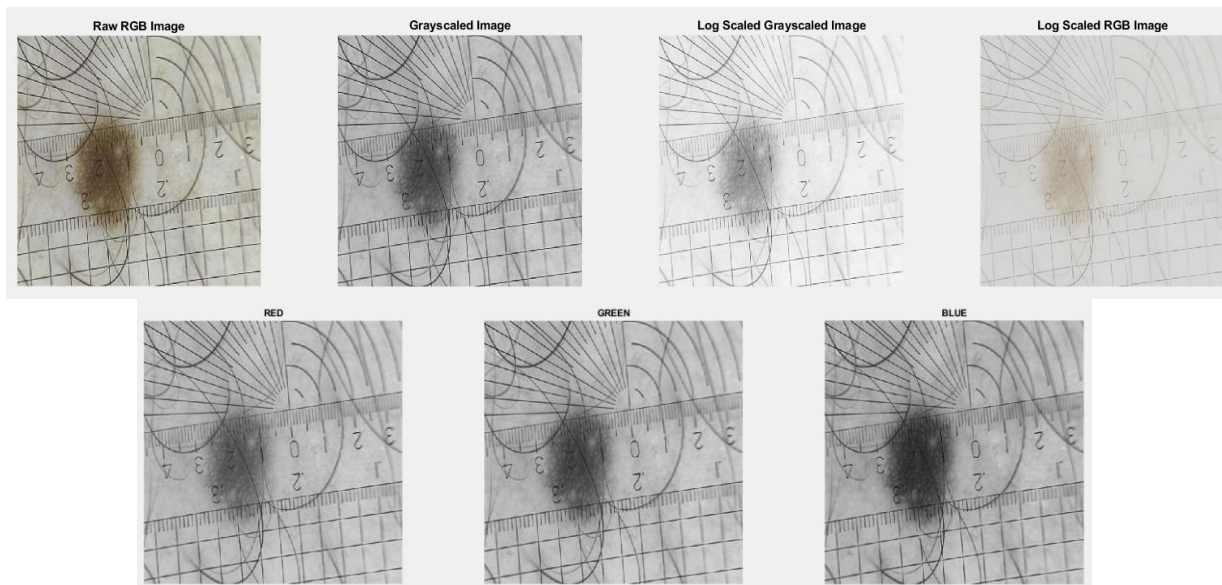


Image Transformations



Current Work: Preprocessing

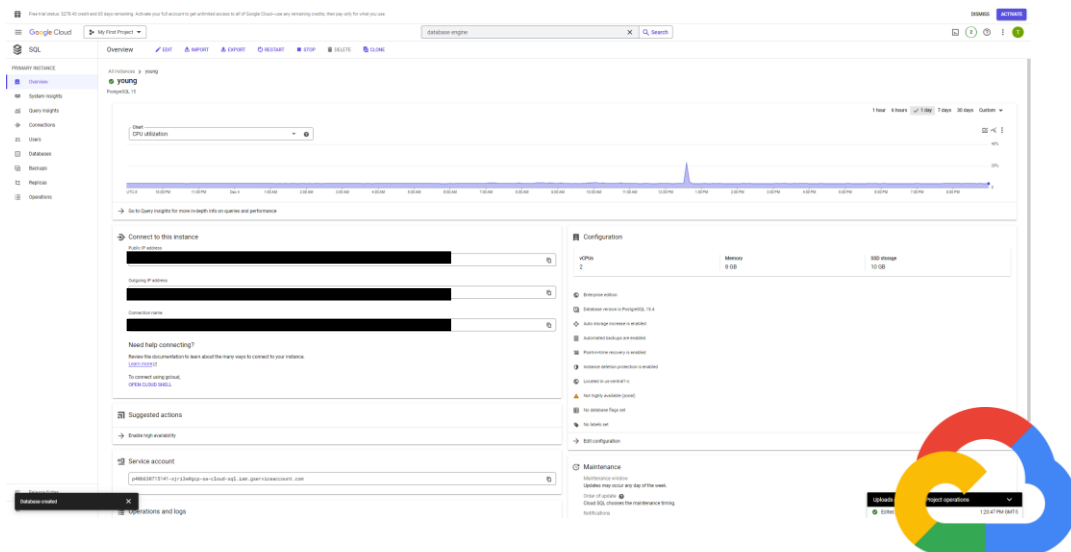
- Image Processing progress currently focused on preprocessing, which seeks to transform images to allow for more information to be revealed for eventual usage in a Machine Learning Algorithm (MLA)

Future Work: Image Segmentation

- Segmentation allows for the isolation of the region of interest and characterization of likewise features, which can be used in later feature extraction and lesion-type characterization for MLA training [3]

Sources: [3]: A. Shetty, K. Shah, M. Ravi, R. M. Sanghvi, S. A. Save, and Y. Patel, "Skin Cancer Detection Using Image Processing: A Review," *Lecture notes in networks and systems*, pp. 103–121, Jan. 2022, doi: https://doi.org/10.1007/978-981-16-6407-6_11.

CLOUD STORAGE WITH GOOGLE CLOUD PLATFORMS

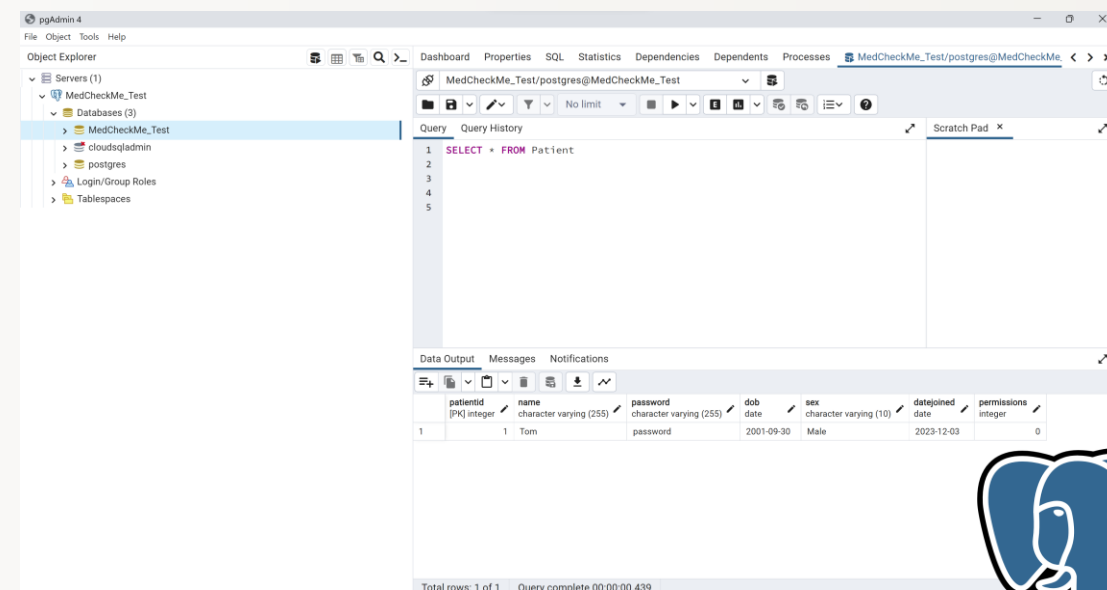


We set up a free trial on Google Cloud platforms to host the database for our application.

- Cost effective free trial
- Sponsor is pursuing relationship with Google



RELATIONAL DATABASE WITH PostgreSQL



After successful prototyping with SQLite, we used Postgre for organizing the data.

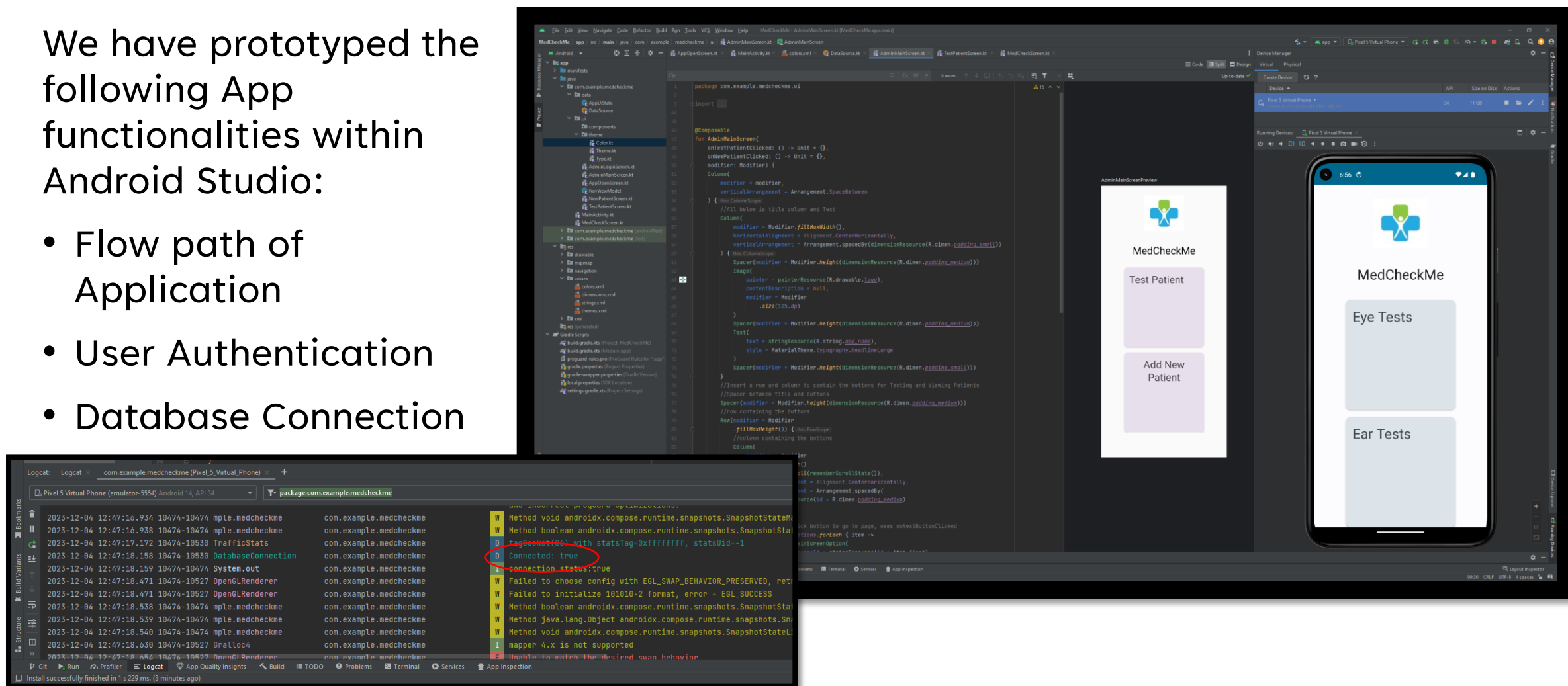
- Well documented and very popular
- Open source → FREE

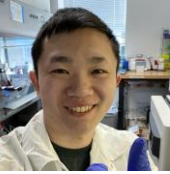


APP IMPLEMENTATION IN ANDROID STUDIO

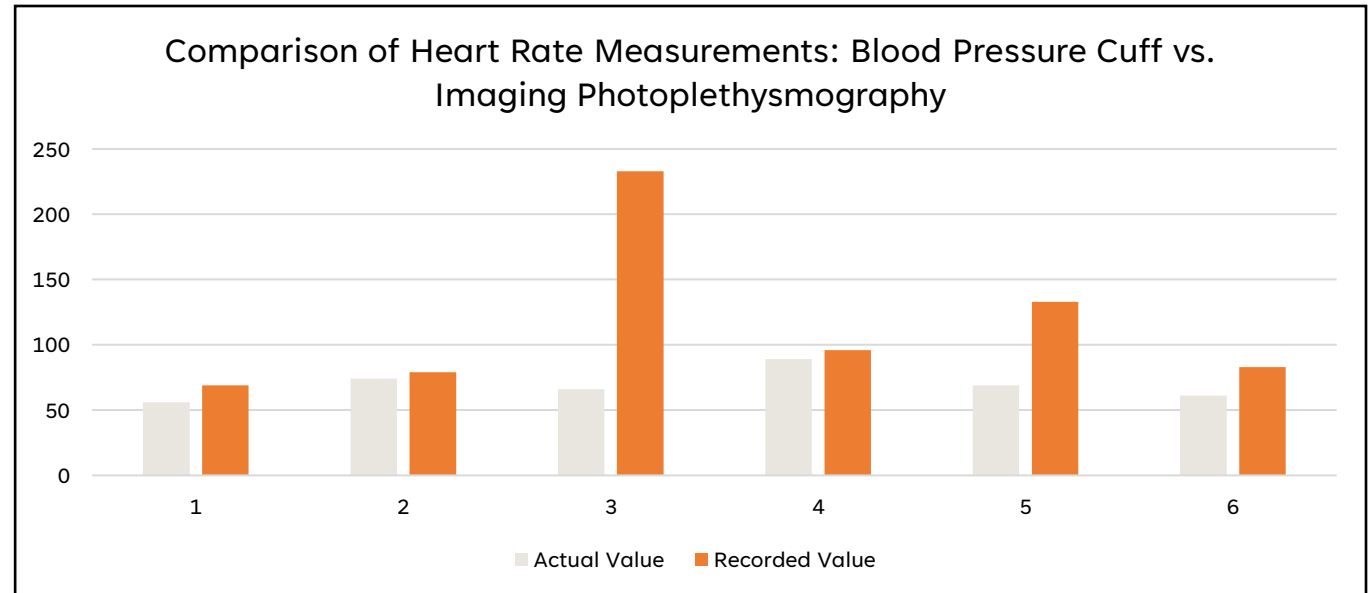
We have prototyped the following App functionalities within Android Studio:

- Flow path of Application
- User Authentication
- Database Connection



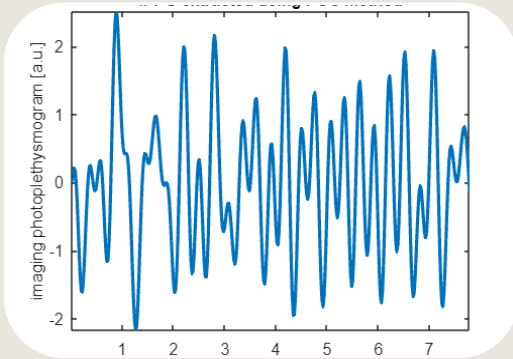


TESTING AND VERIFICATION

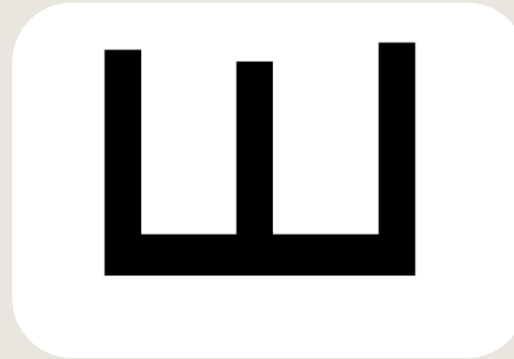


- We tested heart-rate values against an Omron blood pressure cuff, which supplies FDA approved heart-rate values.

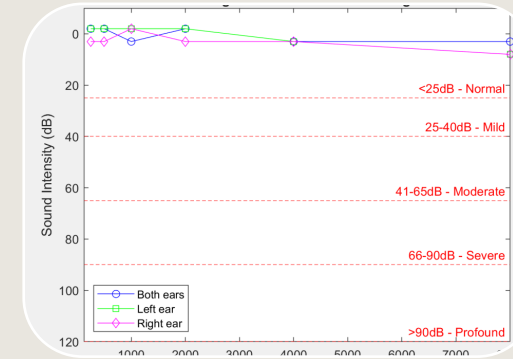
SUMMARY



Heart Rate and Blood Pressure



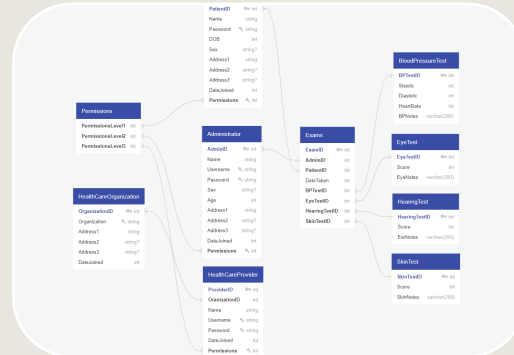
Visual Acuity and Color Blindness Screening



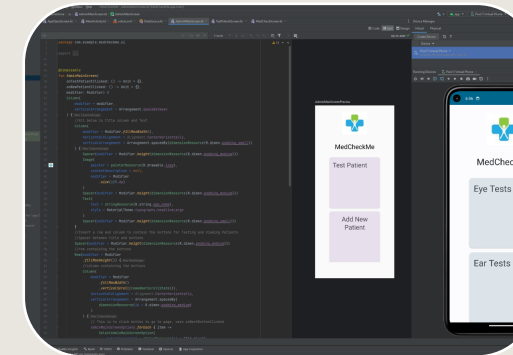
Ear Exam Using Pure Tone Testing



Skin Lesion / Anomaly Documentation



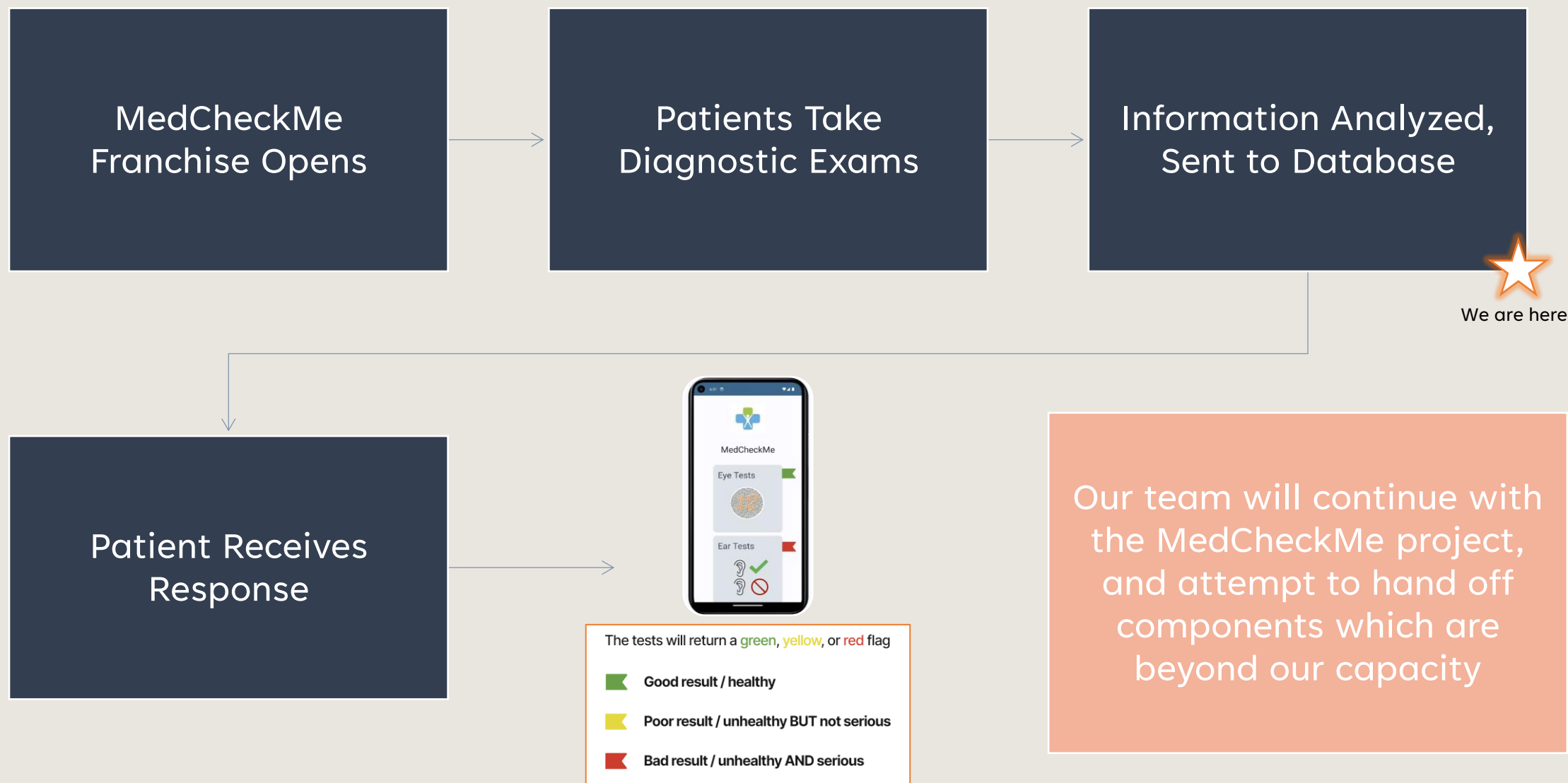
Database Design With Postgres



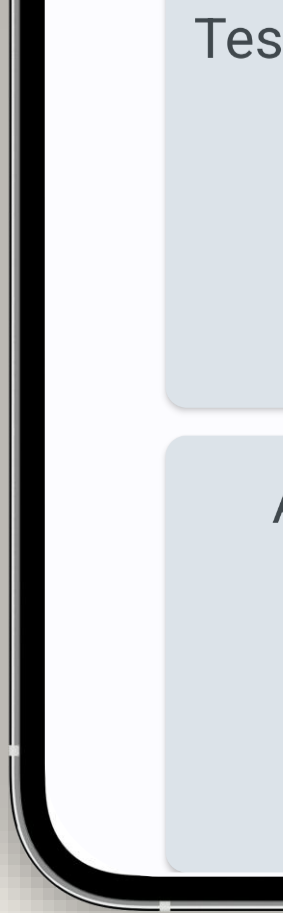
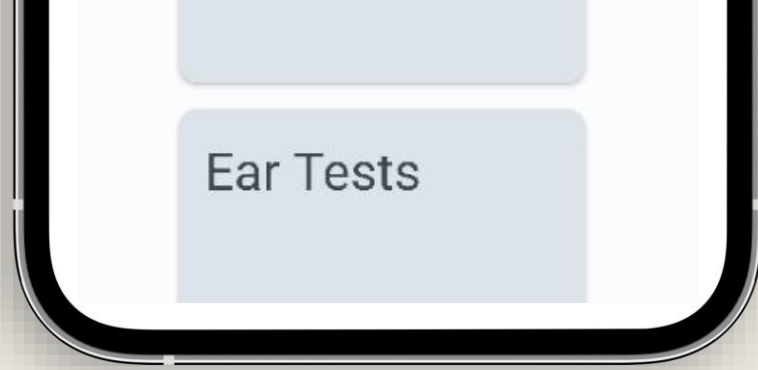
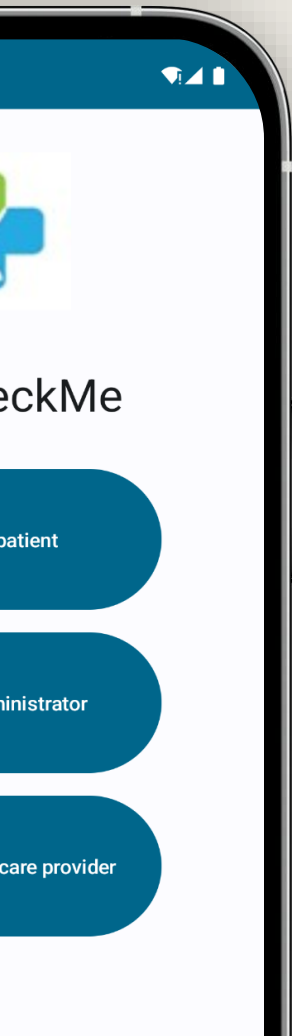
Android Developer Implementation



FUTURE WORK: FLOW-PATH



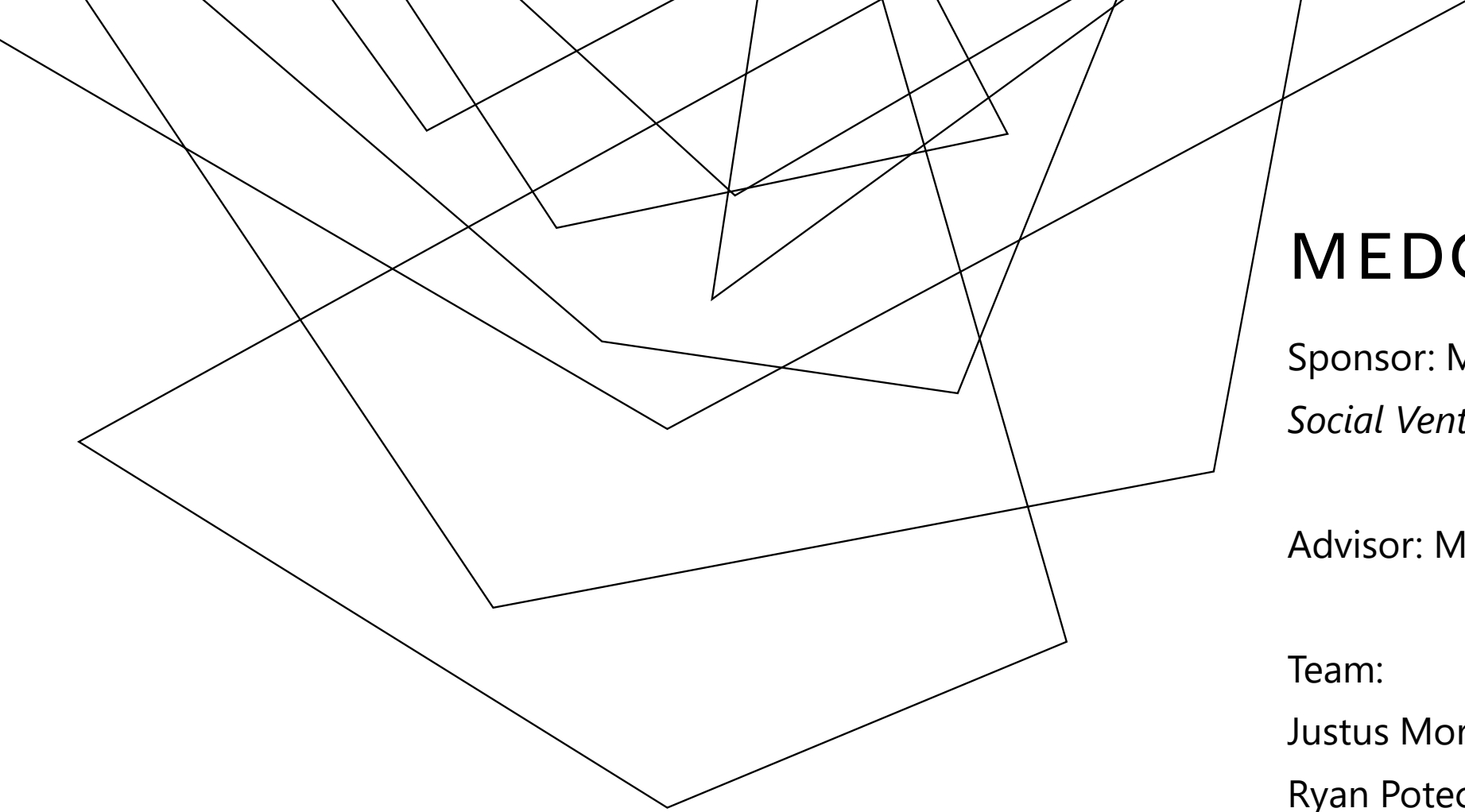
MedCheckMe has **transformative potential** to
address unmet healthcare needs and
empower communities worldwide



THANK YOU

Questions?



A series of thin, black, overlapping geometric lines and polygons that create a complex, abstract pattern on the left side of the slide.

MEDCHECK APP

Sponsor: Marc Blumenthal
Social Ventures Foundation

Advisor: Miguel Mireles-Nunez

Team:

Justus Morales

Ryan Potechin

Sophie Chabot

Samantha Lee

Thomas Young

Trevor Lau

18 October 2023

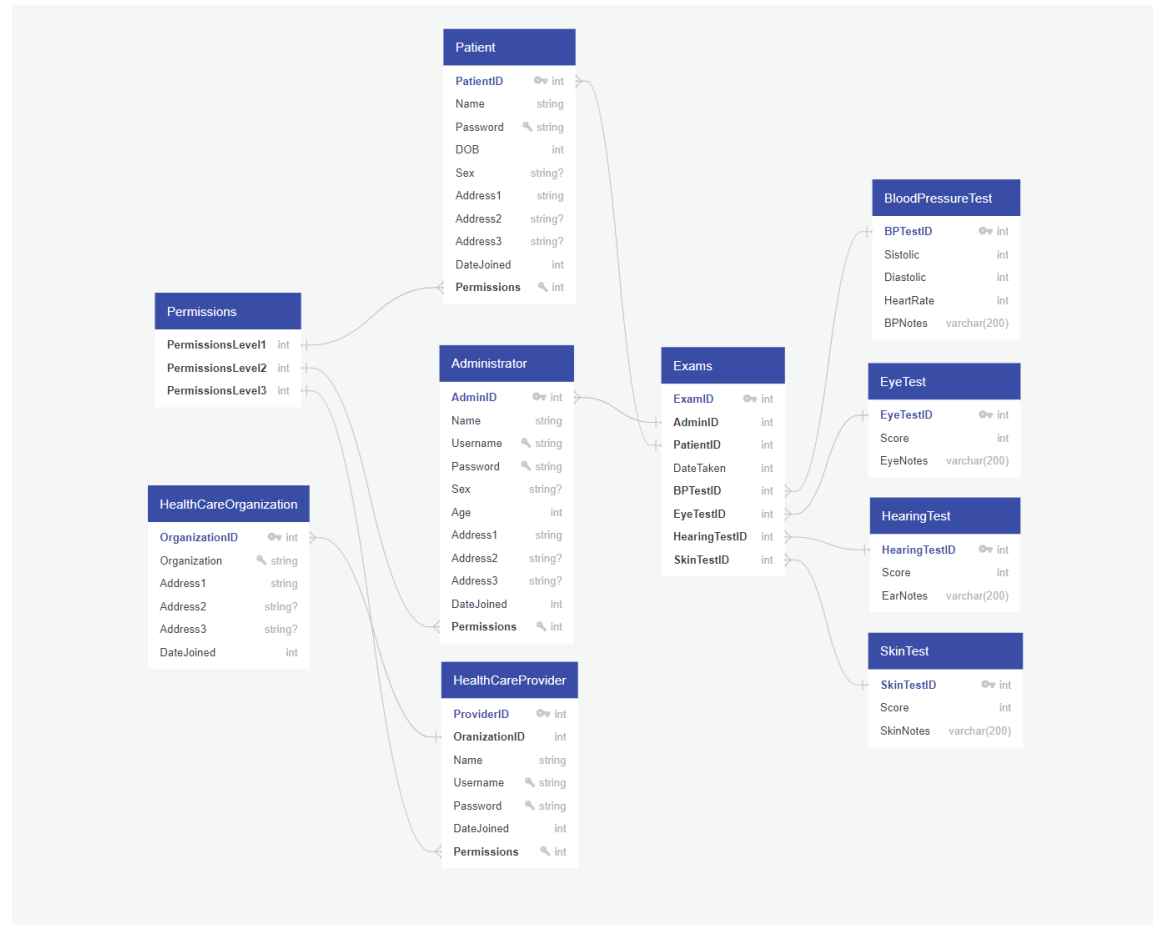
BIOE4792 Bioengineering Capstone 2

APPENDIX A - REVISED PHONE CHART TO REFLECT GOOGLE PARTNERSHIP ASPIRATIONS

Criteria	Cost	Battery Life	Camera	Storage	Relevance	Score
Weight	35	15	25	10	10	
Samsung Galaxy S20	3.61	2.26	2	5	3.72	2.97
Samsung Galaxy S21 5G	2.63	2.53	4	5	3.72	3.17
Samsung Galaxy S22	2.49	2.51	4	5	3.72	3.12
Samsung Galaxy S23	0.85	4.40	5	5	3.72	3.08
ASUS ROG Phone 5S	2.82	3.94	3	5	3.72	3.20
ASUS ROG Phone 6	1.97	4.85	5	5	3.72	3.41
Samsung Galaxy a32	4.00	4.52	3	3	3.72	3.50
Samsung Galaxy A54	3.80	5.00	3	5	3.72	3.70
iPhone 8	4.85	1.90	1	3	0.62	2.60
iPhone X	4.79	1.85	2	3	0.62	2.81
iPhone 11	2.49	4.47	3	3	0.62	2.65
iPhone 12	1.84	3.97	4	3	0.62	2.60
iPhone 13	1.51	2.63	5	5	0.62	2.61
Pixel 7	2.16	3.26	5	5	3.72	3.37
Pixel 6	3.80	3.45	5	5	3.72	3.97
Pixel 5	4.92	3.79	3	5	3.72	3.91
Pixel 4	5.00	1.00	2	3	3.72	3.07

Phone	Cost	Battery Life	Batter Size (mAh)	Camera MP Range	Camera f	Camera sensor (")	Storage (GB)	Relevance
Samsung Galaxy S20	379.99	24:28:00	4000	12 to 64	1.8-2.2	1/2.55-1/1.76	128	74.33
Samsung Galaxy S21 5G	529	25:44:00	4000	12 to 64	1.8-2.2	1/2.55 - 1/1.76	128	74.33
Samsung Galaxy S22	549.99	25:39:00	3700	10 to 50	1.8 - 2.4	1/3.94 to 1/1.56	128	74.33
Samsung Galaxy S23	799.99	34:29:00	3900	10 to 50	1.8 - 2.4	1/3.94 to 1/1.56	128	74.33
ASUS ROG Phone 5S	499.99	32:18:00	6000	5 to 64	1.8 - 2.4	1/1.73	128	74.33
ASUS ROG Phone 6	629.99	36:34:00	6000	5 to 50	1.9 to 2.2	1/1.56	128	74.33
Samsung Galaxy a32	319.99	35:02:00	5000	5 to 64	1.8 to 2.4	1/4.0	64	74.33
Samsung Galaxy A54	349.99	37:16:00	5000	5 to 50	1.8 to 2.4	1/1.56 to 1/3.1	128	74.33
iPhone 8	189.99	22:48:00	1821	12	1.8	1/3.0	64	12.39
iPhone X	199.99	22:33	2716	12	1.8 to 2.4	1/3 to 1/3.4	64	12.39
iPhone 11	549.99	34:47:00	3110	12	1.8 to 2.4	1/2.55 to 1/3.6	64	12.39
iPhone 12	649.99	32:28:00	2815	12	1.6 to 2.4	1/3.6	64	12.39
iPhone 13	699.99	26:13:00	3227	12	1.6 to 2.4	1/3.4	128	12.39
Google Pixel 7	599.99	29:08:00	4355	12 to 50	1.9 to 2.2	1/1.31 to 1/2.2	128	74.33
Google Pixel 6	349.99	30:01:00	4614	12 to 50	1.9 to 2.2	1/1.31 to 1/2.2	128	74.33
Google Pixel 5	179.99	31:36:00	4080	12.2 to 16	1.7 to 2.2	1/2.55	128	74.33
Google Pixel 4	167.5	18:36:00	2800	12.2 to 16	1.7 to 2.4	1/2.55 to 1/3.6	64	74.33

APPENDIX B – DATABASE SCHEMA



- Schema from Quick Database Diagrams Website

APPENDIX C – DIAGNOSTIC EXAMS DECISION MATRIX

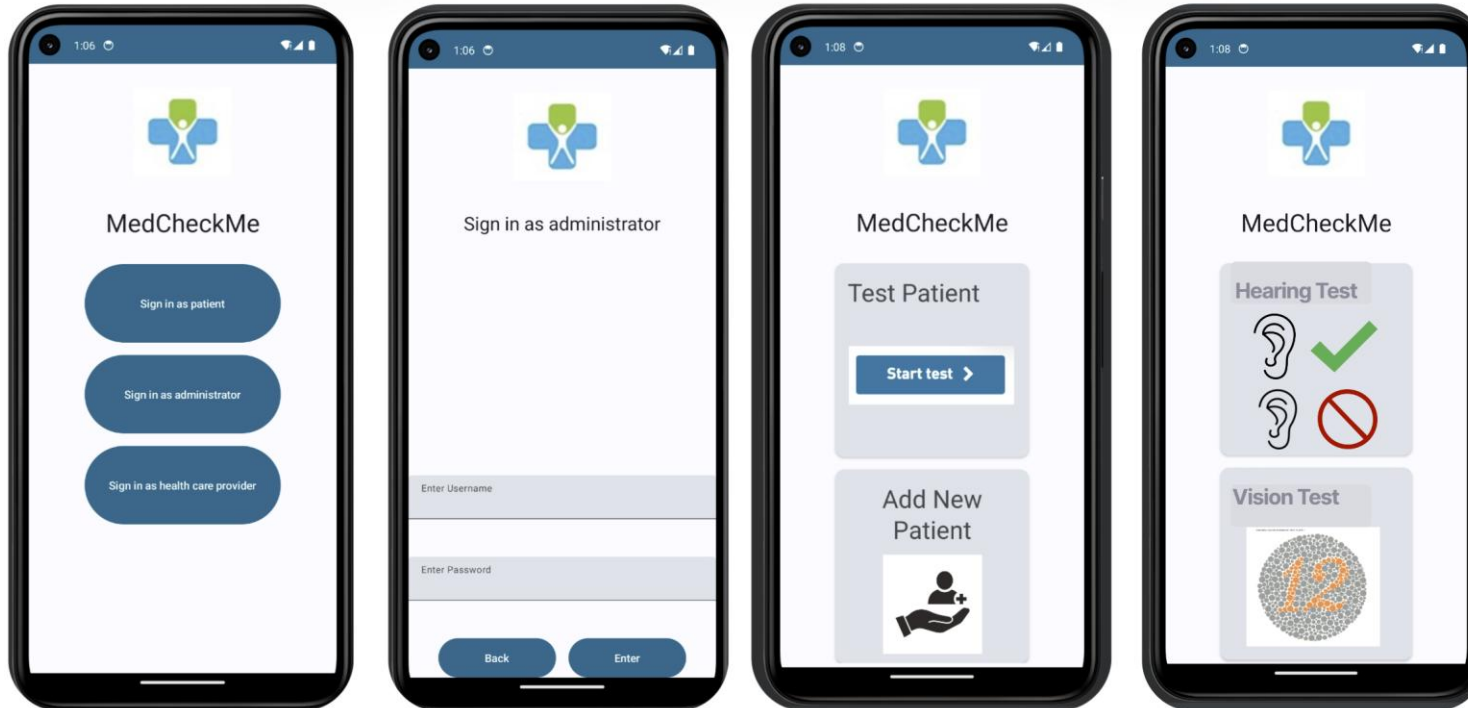
Criteria	Weighting	Blood Pressure	ECG/EKG	General Vision Assessment	Vision Disease Assessment	Hearing Test	Skin	Respiration/Pulse O	Heart Rate and HR
Relevance of Data [1]	40%	4	3	4	3	4	4	2	3
Ease of Measurement [4]	20%	4	3	2	3	4	3	4	4
Accuracy [2]	30%	4	2	3	3	4	3	4	4
Feasibility [3]	10%	2	3	4	1	5	3	3	4
		3.8	2.7	3.3	2.8	4.1	3.4	3.1	3.6

- Decision matrix for diagnostic exams

APPENDIX D – ALL DESIGN REQUIREMENTS

System/Subsystem	Requirement
General System Design Requirements	
	Diagnostic check-ups shall be kept low cost.
	The device shall include pertinent diagnostic tests.
	The device and potential attachments shall be able to be cleaned with medical-grade cleaners without abnormal degradation of function.
	The device and potential attachments shall be able to withstand regional environmental conditions without abnormal degradation of function.
	The app shall have a system for unique patient identification.
	Diagnostic check-ups shall not require internet access.
	The device should have sufficient local storage capacity to perform offline diagnostic check-ups.
	The device shall provide feedback based on diagnostic results.
	The device should provide methods of communicating results with health care professionals.
Eyesight Subsystem	
	The device shall evaluate general vision with visual acuity.
	The device shall evaluate color vision deficiencies with plates composed of varying numbers and colors.
Hearing Subsystem	
	The device shall evaluate hearing loss by playing back tones ranging from 0-120 decibels (dB).
	The device shall evaluate hearing loss by playing back tones ranging from 250-8000 Hertz (Hz) in ascending octaves.
Blood Pressure Subsystem	
	The device shall produce blood pressure measurements with discrepancies less than or equal to 5 mmHg for both systolic and diastolic values when compared to values produced through auscultation.
Heart Rate Subsystem	
	The device shall be able to accurately measure a subject's heart rate in beats per minute with an average error less than or equal to 5% of the overall value.
Skin Testing Subsystem	
	The device shall utilize a magnification lens between 2x and 100x magnification for skin imaging.
	The device shall take and store images on the interface device's integrated hard drive.
Database Subsystem	
	Patient data storage shall adhere to local regulations for Personal Health Information.
	Database shall implement a method of information access control.
	Database shall implement a method of audit logging.
	Database shall implement a method of encrypting data.

APPENDIX E – EXAMPLE OF PHONE FLOW-PATH AS OF NOW



- Phone Flow-path